Oxygen Straight to the Brain: An Overview of Hyperbaric Oxygen Therapy for a Variety of Brain Morbidities

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Abstract

Hyperbaric oxygen therapy as a treatment for conditions like traumatic brain injury, posttraumatic stress disorder, and migraines would seem intuitive, given its effect on condition-related ischemia and inflammation. However, hyperbaric therapeutic impacts for these in acute and chronic, or prolonged symptoms are elusive. This narrative review of hyperbaric's utility provided in sections per disease renders first a review of conventional pathological mechanisms and then articulates hyperbaric treatment targets versus their respective impacts. Multiple challenges exist using hyperbaric oxygen therapy for each morbidity, even in tertiary and adjunctive treatments. An almost universal shortfall across studies includes a lack of consistent, appropriate patient selection criteria intersected with delivery timing of therapy to symptomatic target, necessary to provide a higher fidelity in treatment metrics. Further research into these respective conditions is needed along with a revisitation of hyperbaric oxygen therapy's application to their conventional pathological mechanisms, lending new perspective to their employment and efficacy.

Introduction

Hyperbaric oxygen therapy (HBOT), well known for treating diving-associated injuries, such as decompression sickness and arterial gas embolism, has found favor in treatment of many other medical ailments. HBOT uses 100% oxygen delivered at a pressure greater than one atmosphere for a prescribed period of time depending on the pathology, *e.g.*, as little as 20 min for carbon monoxide poisoning or more than 6 h for arterial gas embolism. The pressure is obtained through the use of a hyperbaric chamber, which can be small enough for a single person or as large as an entire room in order to treat multiple patients at once. Oxygen is introduced through individual masks in a group setting or in the individual chambers, where it is pumped into the entire chamber with titration of gas concentration and pressure levels managed by a physician and technician appropriately trained in hyperbaric operations. Currently, the Food and Drug Administration (FDA) has approved HBOT for decompression sickness, carbon monoxide poisoning, wound treatment, gas gangrene, radiation, and crush injuries, as well as sudden, acute vision, and hearing loss suspected secondary to vascular blockage (Fig. 1) (1).

Instead of covering the widely accepted and known FDA-approved indications for HBOT, the focus of this section article will be some of the lesser known and yet to be proven off-label uses of HBOT that aim to use physiological effects of breathing 100% oxygen at absolute atmospheres (ATA) greater than at sea level to treat traumatic brain injuries, postconcussion syndrome, posttraumatic stress disorder, and migraines.

Traumatic Brain Injury

A traumatic brain injury (TBI) is the result of a direct blow to the head, neck, or body resulting in an impulsive force being transmitted to the brain, initiating a pathological neurotransmitter, and metabolic cascade. In addition, there is possible axonal injury, change in blood flow and inflammation affecting the brain, presenting with immediate or delayed signs and symptoms that commonly resolve within days, but may be prolonged to weeks, months, and in some cases, years (2).

Each TBI is unique in its mechanism of injury, symptom onset, variety of symptoms, and duration of such symptoms. TBI is routinely categorized based on severity as mild, moderate, and severe with an additional classification of mild-complicated, which is noted to be a TBI with mild signs and symptoms,

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Figure 1: The Recompression Chamber at Naval Base Guam Diver Locker (designated RCF-6500), is commonly used for the treatment of decompression sickness (DCS aka "The Bends"), and Arterial Gas Embolism (AGE). It is equipped for hyperbaric administration of oxygen in the treatment of carbon monoxide poisoning, gangrenous tissue, and multiple other diseases provided clearance and authorization are granted (photo courtesy of Dr. Edgar VanHorne, 2020, personal collection).

but abnormal findings on neuroimaging, such as CT. The differentiation between mild, moderate, and severe are predominately categorized based on initial symptoms at time of the head injury according to the Glasgow Coma Scale (GCS) with mild having a score of ≥13, while moderate is diagnosed with a score range of 9 to 12, and a severe TBI is noted for any GCS score ≤ 8. Current recommendations for concussion treatment involve initial rest, followed soon thereafter by activities of daily living, mild aerobic exercise, and a gradual progression into the individual's normal life as it was prior to the injury (2). But what if the symptoms are prolonged beyond the normal resolution duration of a month? What if the symptoms last months or even years? It is the prolonged symptoms, which include headaches, dizziness, difficulty concentrating, and overall cognitive impairment lasting greater than 3 months, that are diagnosed as postconcussion syndrome (3). These symptoms, at this duration, are the most difficult to understand and the most frustrating to treat. Some patients learn to live with the symptoms, while others seek out alternative treatments to include acupuncture, nerve blocks, or the use of HBOT (Fig. 2).

The hypothesis behind the use of HBOT in the treatment of TBI is to treat the secondary pathological mechanisms associated with the concussion: ischemia and inflammation (4). Henry's Law states that at a constant temperature, the amount of a given gas that dissolves in a liquid is directly proportional to the partial pressure of that gas in equilibrium with that liquid. Therefore, the goal of HBOT is to, at atmospheric pressures greater than one ATA, increase the availability of oxygen within the bloodstream above the hemoglobin saturation capacity. Which, in turn, promotes the healing of brain tissue from the ischemic damage as a result of the blood flow change, while also reducing the inflammation that directly affects the brain. Hu et al. (5) postulated that traumatic brain injury is characterized by cytokine release, neutrophil activation, and microvascular adherence. To counter these effects HBOT reduces inflammation via increasing a multitude of anti-inflammatory cytokines, such as interleukin-10, while decreasing a variety of proinflammatory factors to include TNF-alpha.

Testing the hypothesis that HBOT can be used in the treatment of TBI has taken many different forms with a variety of treatment tables and durations with the most advantageous results coming from treatment pressures taking place at 1.3 to 1.5 ATA for 60 min daily for 2 to 4 wk (4,6). On the other hand, the least effective pressure is found to be 2.4 ATA, with pressures greater than 2.0 ATA proving to be much less efficacious at the treatment of mild TBI postconcussion syndrome symptoms to the point that the cost and risk greatly outweighs the benefit.

Although studies in the past have shown little benefit in treating postconcussion syndrome with HBOT, one might find limitations in these studies secondary to the subjective nature of their reporting of symptoms. Recent advancements in technology have proven benefit in studying traumatic brain injury and postconcussion syndrome, with brain imaging via single-photon emission computed tomography (SPECT) and perfusion sequences during MRI evaluating cerebral blood flow and brain metabolism leading the way. In 2022, Harch et al. (6) performed a systematic review noting the correlation between specific cognitive function improvements with metabolic brain imaging changes after HBOT noted in the perirhinal cortex, prefrontal cortex (inferior frontal gyrus), and anterior cingulate gyrus. Perirhinal cortex change after HBOT was most significant with memory improvement. Prefrontal cortex-inferior frontal gyrus change after HBOT were prominent in all patients with executive function improvement. Anterior cingulate gyrus with HBOT noted improvement with attention (6).

In a more specific hypothesis, it is thought that HBOT may promote neurogenesis of endogenous neural stem cells by supplying an excess of oxygen and energy used by the underlying oxygen-dependent regeneration processes (4). Therefore, through HBOT, improving and preserving the newly generated neuronal functioning that was previously damaged is the goal. In their retrospective analysis, Hadanny et al. (4) correlated between specific cognitive function improvements with the metabolic brain imaging changes noted on SPECT which in turn

Figure 2: The main entrance hatch that leads to the interior of the recompression chamber. It has room for one patient bed (right), and for an inside tender/dive medical technician (seated left) — who renders care and oversees patient monitoring during treatments (photo courtesy of Dr. Edgar VanHorne, 2020, personal collection).

give further strength to understanding the brain's functionality with the correlation between certain neurological deficits prior to and post HBOT. The following associations were made with specific gains after HBOT according to Hadanny's analysis:

1. Patients with significant memory deficits noted an improvement in cognition after HBOT, which was associated with and noted on SPECT with an increase in perirhinal cortex activation.
2. The inferior frontal gyrus activity was pronounced after HBOT in individuals with significant executive function deficits.
3. The anterior cingulate gyrus, used predominantly for error detection and attention, was noted to increase in activation after HBOT on SPECT and also was subjectively noted to improve overall attention capabilities.

Recent findings support the notion that HBOT can and should be considered for patients with chronic neurocognitive deficits following TBI (4,6). On the other hand, there needs to be more detailed, standardized research regarding pressures, frequency, duration, and efficacy prior to an individual risking injury (barotrauma, oxygen toxicity, worsening cataracts) or spending the time and money (average out of pocket cost for monoplace chamber in Southern California is $160 to $200/h with "recommendation" of 40 sessions versus at academic center, each session can cost $1700) on a treatment that may not resolve their symptoms.

Posttraumatic Stress Disorder

Posttraumatic stress disorder (PTSD) is a complex, chronic, severe, and disabling psychiatric condition that may develop in response to actual or threatened physical or psychological trauma (7). PTSD is characterized by intrusive thoughts, nightmares, flashbacks of traumatic events, avoidance of reminders of trauma, hypervigilance, improper stress response, and sleep disturbances, leading to social, occupational, and interpersonal dysfunction. Although the pathophysiology of PTSD is not well understood, studies using magnetic resonance imaging (MRI) have demonstrated decreased volumes in the prefrontal cortex and limbic system, notably in the hippocampus, amygdala, and anterior cingulate cortex, which possibly accounts for the dysregulation of emotions and memory impairment seen in patients with PTSD (8,9).

In the wake of military conflicts in Iraq and Afghanistan, much attention and resources have been dedicated to the study of TBI and PTSD, both epidemics among the veteran population. Several studies have explored the possible utility of HBOT in the treatment of veterans with TBI with or without comorbid PTSD, most of which have demonstrated clinical improvement in PTSD symptoms (10). Notably, a small phase I study, demonstrated that low-pressure HBOT (40 sessions, 60 min each, breathing 100% at 1.5 ATA) for veterans with a history of PTSD in the setting of TBI was generally safe and well tolerated (11). Furthermore, the study yielded statistically significant improvement in PTSD symptoms, with reduction in the mean PTSD Checklist-Military (PCL-M) score of over 20 points (on a scale of 17 to 85), or approximately 30% (11). However, it should be noted that despite the significant improvement on average, many of the subjects continued to meet diagnostic criteria for PTSD, and the study may have been confounded by the presence of psychiatric disorders, most notably depression and anxiety, as well as polypharmacy (12). In a prospective case-control pilot study of United States military veterans with TBI and PTSD, Harch et al. (13) found that immediately after undergoing a 4-wk course of HBOT (40 sessions, 60 min each, breathing 100% oxygen at 1.5 ATA), of their 23 subjects previously diagnosed with PTSD, 12 (52%) no longer met diagnostic criteria for PTSD and showed a statistically significant reduction in mean PCL-M score of 18.8 points from baseline. The more recent BIMA (brain injury and mechanisms of action of HBO_2 for persistent postconcussive symptoms after mild TBI) study found that while PTSD symptoms tended to improve immediately posttreatment, these results were not sustained long-term (14). This phase II, randomized, double-blind, sham-controlled trial explored HBOT for military personnel with history of mild TBI, including those with comorbid PTSD, using a 12-wk, 40-session HBOT protocol. At week 13, subjects with baseline diagnosis of PTSD who underwent HBOT showed statistically significant changes in PTSD Checklist-Civilian (PCL-C) scores (–9.3) compared with sham (3.0), for a difference of –12.3. However, improvement did not persist at follow-up 6 and 12 months posttreatment (Fig. 3).

While investigations of HBOT as a potential therapy for PTSD have tended to include subjects with comorbid TBI, a comprehensive review of the literature returned only a single study to date exploring HBOT as a possible adjunct to pharmacotherapy and psychotherapy for the treatment of PTSD in isolation (15). Doenyas-Barak et al. enrolled Israeli combat veterans with combat-associated, treatment-resistant PTSD persisting for 4 years or more into a randomized, prospective controlled trial to assess the efficacy of 60 sessions of HBOT on PTSD symptomatology, as well as changes in brain microstructure noted with functional imaging (13). The primary clinical outcome was assessed using the Clinician-Administered PTSD Scale for DSM-V (CAPS-V), which measured a baseline score, at 1 and 4 wk postprotocol. The authors observed that subjects who underwent HBOT demonstrated statistically significant reductions in CAPS-V score of almost 39%, while the

Figure 3: A comparison of the Built-in-Breathing System (BIBS) regulator mask typically worn by the Inside Tender (left) and a Diving Hood (with fitted plastic Neck Dam) typically worn by patients (right). This system enables divers and patients alike to inhale breathing mixtures different from the internal chamber atmosphere, enabling recompression chamber operations requiring breathing air, nitrogen-oxygen, helium-oxygen, or 100% oxygen-rich mixtures (photo courtesy of Dr. Edgar VanHorne, 2020, personal collection).

control group did not show any statistically significant change (15). Diffusion tensor imaging (DTI) MRI also showed improvements in brain microstructure integrity in the frontolimbic circuit for the treatment group, particularly in the following regions: anterior limb of the left internal capsule, right corona radiata, the genu of the corpus callosum, the parietal white matter, the cerebral peduncle, and the fornix (15). Although functional MRI (fMRI) did not show any significant changes for whole-brain task-related activation from baseline to posttreatment, fMRI did demonstrate improved activity in several brain structures, including the left dorsolateral prefrontal cortex, middle temporal, and temporal gyri, as well as both thalami, the left hippocampus, and the left insula (15). DTI MRI results point toward restoration of the frontolimbic circuit, which may account for improvement in CAPS scores. This increase in CAPS scores correlates with a decrease in PTSD symptomatology, particularly emotion and memory (15). Of note, studies emphasized the importance of the hippocampus, identified as a significant structure affected in PTSD. The hippocampus' role in memory and fear conditioning is appreciated through an increase in hippocampal activity on fMRI and microstructural improvement in the fornix, the major output tract of the hippocampus, on DTI MRI (15). Therefore, the association between the restoration of hippocampal activity status-post HBOT and the improvement in symptoms as seen in previous studies may be loosely linked (16,17).

An interesting and unexpected outcome of the Israeli study was the surfacing of previously inaccessible memories, a finding which was neither a predefined endpoint nor solicited by the investigators, but rather voluntarily disclosed by several subjects (15). These memories usually surfaced insidiously after 25 to 35 HBOT sessions, although one participant reported an abrupt flashback after the fifth session. While these memories tended to cause severe distress initially, integration of the memory and resolution of the distress generally followed. Interestingly enough, similar findings also have been observed in studies of HBOT for the treatment of fibromyalgia, with subjects reporting resurfacing of repressed memories of childhood sexual abuse (18). Furthermore, deep brain stimulation has been shown to improve hippocampal neurogenesis in animal models, as well as surfacing of repressed memories in humans, while studies of the effect of HBOT on hippocampal cells in culture have been shown to induce orthodromic conduction and neuroplasticity (19,20). Given increases in hippocampal activity on fMRI in the Israeli study and the suspected role of the hippocampus in PTSD, surfacing of memories among test subjects may provide further evidence of HBOT's effect on hippocampal neuroplasticity, and its potential relevance in the treatment of PTSD (15).

Unfortunately, studies exploring HBOT as a treatment for PTSD have been limited by small sample sizes and poor follow-up and confounded by comorbidities to include TBI and polypharmacy, and probably underlying preexisting depression and anxiety (11,12). And while the available data reliably show favorable outcomes in symptomatology and imaging immediately posttreatment, at least one study failed to demonstrate persistent improvement 6 and 12 months later, so the plausibility of HBOT as long-term treatment for PTSD is less clear (11,13–15). Perhaps, further studies could investigate the potential of follow-up HBOT sessions for continued treatment of PTSD, particularly as compared with medical management. HBOT does appear to be well tolerated among patients with PTSD, with the most common complication being mild, reversible middle ear barotrauma, often in the setting of upper respiratory infection. Given the relatively benign nature of HBOT, it warrants further investigation as a potential adjunct in the management of PTSD.

HBOT and Migraines

In 1987, with great contention, Gabb and Robin stated that "Hyperbaric oxygen therapy is a treatment in search of a disease" (21). Current evidence proves this statement to be truer now, more than ever before, regarding the treatment of migraines with HBOT. A broader understanding of pathogenic mechanisms and newly articulated therapeutic targets, combined with rapidly expanding apertures of therapy, particularly from clinical applications in China, lend new potential pathways for the hyperbaric and normobaric approach to this neurological conundrum (Fig. 4) (22,23).

Migraines are a widely encountered, debilitating neurological disorder, and although the symptoms are well defined by either patients who can describe a detailed symptomatic spread and sequential deficits or by vascular and cortical activity disturbance on brain imaging, the mechanisms behind the more debilitating symptoms are variable and not well understood. Traditionally, migraine pathophysiology has been articulated in an almost dichotomous split between electro-sensory disturbance versus vascular insult (24–27). The sensory processing description is borne out of an electrical disturbance; that the brain's electrical signals are not transmitted properly (producing visual aura versus olfactory or sensory changes from triggers), leading to an unusual and disorganized wave of brain activity, therein producing symptoms like photophobia, phonophobia, and visual changes with or without aura (28). Alternatively, and typically more acutely, migraines have been viewed through the lens as a vascular insult, and more specifically a vasoconstrictive versus vasoocclusive event, resulting

Figure 4: Located at the opposite end of the chamber is the door to the "Outer Lock," a smaller isolable chamber which can be separately decompressed and brought to the surface, while the "Inner Lock" housing the patient remains at the same compression depth in continuance of the treatment. Used to change out medical personnel or bring aboard additional equipment during therapy, it also contains a "Med Lock" (left; a small hatch compartment approximately 12 inches in diameter) through which IV medications, or other fluids may be passed to/from inside (photo courtesy of Dr. Edgar VanHorne, 2020, personal collection).

in ischemic versus trigeminal pathway insult resulting in nociceptive and sensory symptoms (29,30).

When considering more chronic migraine mechanisms, which are not fully understood, they are thought to be more likely caused by inflammatory responses resulting in neuropeptide release, spurning the cascade of inflammatory cytokines (31). More specifically, short-chain neuropeptides involved in pain signal transmission like Substance-P and calcitonin gene-related peptide, among others, which have been suggested to play critical roles in the initial inflammatory genesis of migraine episodes (32,33).

In a recent take on this traditional mechanistic difference, Pourshoghi et al. have described both the well-known vascular theory behind migraines (*e.g.*, vasodilation of meningeal vessels irritating trigeminal nerves) and the lesser-known Neural Theory of migraines, describing cortical hyperexcitability via cortical spreading depression (CSD) (34). Further described by Lauritzen and Hadjikhani, CSD depolarization waves are associated with failure of brain ion homeostasis, an efflux of excitatory amino acids from nerve cells, an increase in energy metabolism, and changes in cerebral blood flow. The same authors state that there is strong clinical and experimental evidence suggesting that CSD is involved in the mechanism of migraine, stroke, subarachnoid hemorrhage, and traumatic brain injury (35). Lauritzen and Hadjikhani articulate that brain imaging involving cerebral blood flow and patient-specific migraine attacks showed what was termed a "spreading oligemia," a wave of reduced blood flow that propagated across the brain at the same rate and with the same signs of vascular impairment as CSD (35,36). As a result, it is intriguing to consider that the interweaving of brain perfusion, the consequential impact on brain-energy supply, and hemodynamic disturbances that occur with transit of ionic and metabolic disequilibria resulting in neurological symptoms from vulnerable nerve cells could all be potentially amenable to HBOT, therefore providing new pathophysiologic mechanisms and therapeutic targets upon which to intervene.

Given these symptomatic pathways, potential migraine mechanisms lend themselves to therapeutic intervention by HBOT or normobaric oxygen therapy (NBOT) at sea level via the known benefits from increased oxygen-hemoglobin binding, increased oxygen dissolution in plasma, improved oxidation balance, reactive vasoconstriction, and improved collateral circulation. NBOT is the use of increased levels of oxygen concentration (approximately 40%) delivered via nasal cannula, facemask, or within an enclosed space that functions within normal atmospheric pressures, but at an increased oxygen saturation level. NBOT has demonstrated notable success in the treatment of cluster headaches via oxygen delivered at higher concentrations to circumvent vasoconstrictive restrictions through saturation of both red blood cells and plasma (37,38). With articulated oxygen-induced reflexes causing constriction of distal cerebral resistance vessels, NBOT's ability to maintain subsequent oxygenation demonstrated theoretical promise for HBOT (39,40). In addition, HBOT's therapeutic effect through oxygen as a serotonergic agonist and an immunomodulator of cytokines, neuropeptides, and Substance P is critical in addressing more of the inflammatory and neural pathway pathogenic methodologies (41,42). Unfortunately, the review of HBOT and NBOT as a migraine prophylactic therapy has been less promising.

The current evidence surrounding NBOT and HBOT efficacy for migraines is not great after the review of five main studies, two of which have low-double digit meta-analysis and one Cochrane review of 11 studies, which Bennett et al. (43) has been the historical center regarding this substantiation. These studies collectively state that the therapeutic modality is plagued by issues in true randomization methods versus allocation concealment methods, and when considering the standardization of therapy provided, the quality of the evidence was low (44). NBOT historically has demonstrated termination of acute headaches for both migraines and cluster headaches, but consistency has proven elusive. HBOT does not seem effective for cluster headaches, nor does it show good preventive or acute symptom management other than pain (*i.e.*, nausea/vomiting). Neither HBOT nor NBOT reduces the requirement for rescue medication, nor do they provide consistent prophylaxis or reduce the need for abortive medications (45–47). There are three studies looking at more chronic therapy, which in preliminary reports, are demonstrating improvements in reduction of overall attacks, frequency, and duration of episodes, although subjective measures are withstanding (48–50). As a result, when approaching the therapeutic target of underlying inflammatory pathophysiology in migraines, HBOT for migraines does not appear to have any benefit in long term treatment and as result cannot be supported medically, nor should it be financially, as the risks outweigh the benefits.

Despite all this, NBOT and HBOT merits another evaluation of its therapeutic efficacy in migraines for several reasons. First, it is a nonpharmacologic answer that has little resistance, is noninvasive, and may offer an improved side effect profile when compared with typically used triptans and ergotamines. Second, both NBOT and HBOT modalities stand to benefit from expanding the indications to include the various classifications of migraines. Due to HBOT's potential for developing oxygen toxicity levels, more uniform and standardized treatment guidelines and parameters in HBOT are essential to producing a more consistent protocol for therapeutic comparison in future studies, which continues to plague multiple studies given the

Figure 5: The Divers Control Panel, where the Recompression Chamber Team controls depth, travel rate, treatment time, and breathing mixture, all while simultaneously maintaining video monitoring of the chamber occupants. Video/visual monitoring with voice communications ensure that everyone from the chamber supervisor to the dive medical officer are enabled to keep up with the surge of activity typically encountered during treatments (photo courtesy of Dr. Edgar VanHorne, 2020, personal collection).

wide variety of approach outside of those already articulated as in United States Navy standard treatment tables. Lastly, while HBOT is widely regarded as relatively benign, mitigation of minor complications like barotrauma and the technical barriers surrounding sophisticated equipment versus technological accessibility can be addressed to expand both therapy and subsequent efficacy measures to a wider audience (Fig. 5) (22,23,45).

Conclusions

HBOT has growth potential in terms of treatment for brain injuries and neurocognitive dysfunction including the treatment of TBI, PTSD, and migraines. Unfortunately, multiple shortfalls are identified, such as nonstandardized therapeutic regimens, wide variability in subjective symptoms as reported by the patient, and unclear targets of therapy or articulated symptomatic reduction. While improved articulation of pathophysiological mechanisms and advancements in imaging capabilities may provide new targets of therapy and wider inclusions of indications can expand and propel HBOT's future, we are left with an off market, costly, and complex intervention whose potential benefits are likely to not be covered by insurance until a more standardized approach with definitive objective data emerges. These factors hinder HBOT from considerations at the forefront of neurocognitive treatment regimens at this time.

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The authors are undersea medical officers in the United States Navy, trained in the use of hyperbaric oxygen therapy for its primary role in the treatment of decompression sickness and arterial gas embolism. The views and statements are of their own research and opinions and do not reflect the stance of the Defense Health Agency, Bureau of Medicine, Special Operations Command, or the United States Navy.

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